



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech.(CSE) (CBCS) Degree Examinations

January 2023

22CSE101 – WIRELESS NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Demonstrate supporting diagram and explain small scale and large scale fading in wireless communications.	6	L*2	1	1
	b) Give generic 802.11 frame with frame control field structure and give significance of each field.	8	L3	1	2
	c) List out four different inter frame spaces giving their significances and showing their inter frame space relationship.	6	L2	1	2
2.	a) Outline any three wireless network services and explain in brief giving three properties of the same.	6	L2	1	1
	b) With diagram explain translating frames between wirelesses to wired medium in bridges.	8	L3	1	2
	c) With supporting figure, explain RTS/CTS lockout with and without fragmentation in wireless data transmission.	6	L2	1	2
Unit – II					
3.	a) What is a PS-Poll frame? With frame format explain the fields of PS-Poll frames.	6	L2	2	2
	b) Give generic structure of a 802.11 data frame. Describe the relationship of addressing and DS bits in 802.11 frame and show the Use of the address fields in infrastructure networks.	8	L3	2	1
	c) With diagram explain in brief probe request and probe response management frames.	6	L2	2	2
4.	a) Demonstrate the structure of ACK frame with explanation of every fields.	6	L2	2	1
	b) With necessary sketch explain generic structure of RTS and CTS frames and give the timing diagram of an Uninterrupted frame transmission using RTS and CTS without fragmentation.	8	L3	2	2
	c) Write a note on supported rate information elements with supporting diagram.	6	L2	2	2
Unit – III					
5.	a) With diagram explain 802.1x port authentication architecture.	6	L2	3	1
	b) Discuss the sequence of events in TKIP data transmission. Explain TKIP data encapsulation process with diagram.	8	L3	3	2
	c) List out common 6 EAP methods for authentication with brief description.	6	L2	3	2
6.	a) Show the EAPOL frame format on 802.11 and Ethernet giving details of fields.	6	L2	3	1
	b) Explain with suitable diagram how CCMP support for encryption and integrity protection as part of same process, also show CCMP encapsulation.	8	L3	3	2
	c) Give 5 differences between TKIP and WEP.	6	L2	3	2

P.T.O.

Unit – IV

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|-------|--|---|----|---|---|
| 7. a) | With suitable example explain frequency hopping transmission, showing orthogonal hopping sequence. | 6 | L2 | 4 | 2 |
| b) | Explain method of Encoding a letter D in GFSK and show the encoding using 2 level and 4 levels. | 8 | L3 | 4 | 2 |
| c) | With diagram explain FH PLCP framing and processing. | 6 | L2 | 4 | 1 |
| 8. a) | With suitable example explain the technique behind Direct sequence spread spectrum transmission. | 6 | L2 | 4 | 2 |
| b) | Explain method of Encoding a letter D in DBPSK and show the encoding using DBPSK and DQPSKs. | 8 | L3 | 4 | 2 |
| c) | With diagram explain DS PLCP framing and processing. | 6 | L2 | 4 | 1 |

Unit – V

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|--------|--|---|----|---|---|
| 9. a) | What is a handoffs?. Why handoff is required? Explain issues to be addressed in handoffs algorithm in a cellular networks. | 6 | L2 | 5 | 1 |
| b) | With diagram Explain in detail ETSI broad band access network system linking different high performance radio LANs. | 8 | L3 | 5 | 2 |
| c) | With figure explain concept of sectorization for capacity enhancement in cellular network. | 6 | L2 | 5 | 1 |
| 10. a) | Explain the WAP Client server model with supporting diagram. | 6 | L2 | 5 | 1 |
| b) | With supporting diagram explain the routing in MobileIP. | 8 | L3 | 5 | 2 |
| c) | Write a note on TCP over wireless. Show Classification of TCP over wireless. | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
